Generating Images of Surface Landmines: A Class-Conditional GAN Trained on Surrogate Mines

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Overview

In this project I trained a generative adversarial network (GAN) in PyTorch to generate 64 × 64-pixel colour images of two kinds of surface landmine, the PFM-1 and the PMA-2, on request. A GAN pairs two networks: a generator that turns random numbers into images, and a discriminator that learns to tell its images from real ones, each improving against the other (Goodfellow et al., 2014). The one I built is class-conditional: the mine type is given to both networks as an extra input (Mirza & Osindero, 2014). It learned from crops cut around the labelled items in SULAND, a public set of video frames of mine surrogates lying in grass, leaves and gravel (Vivoli et al., 2024).

In my work in explosive ordnance disposal (EOD) and humanitarian mine action, generated mine images could serve as extra training images for a model that finds mines in drone or survey photos, as recognition training and awareness material, and to build digital worlds for drones, or drone operators, to train in. What matters in those uses is the shape of the item rather than its colour, since mines can be painted to camouflage, and a picture that teaches people what a mine looks like should be grounded in engineering data and be as accurate as possible. I therefore judged the generated images in four ways tied to these uses: whether they copy their training images; whether a classifier trained on them works on real photos, at the training site and at a new one; what they look like between the two types; and, in a blind review, whether I could tell them from real crops and whether their shapes are right.

The generator learned both types without copying its training crops, and a classifier trained on real crops recognises most of its images as the type requested. But the images do not

stand in for real ones. A classifier trained on them finds almost none of the real mines, while scoring almost perfectly on generated test images, and in the blind review I told generated crops from real ones in 30 of 32 cases and judged the shape right in only 4 of the 16 generated crops.

Dataset Description

SULAND and Its Labels

SULAND (SUrface LANDmine) was released under a CC BY 4.0 licence by Emanuele Vivoli, Marco Bertini and Lorenzo Capineri of the Media Integration and Communication Center of the University of Florence, with a paper on detecting surface landmines in optical images (Vivoli et al., 2024). Its 33,771 frames, all 1440 × 1920 pixels, come from 55 videos taken with a phone camera of surrogates of the PFM-1 and the PMA-2. The Italian videos are split into training (22,756 frames from 34 videos), validation (2,836; 5) and test (3,743; 6) sets, made by video so that the frames of one video stay in one set, and a separate set of 4,436 frames from 10 videos was filmed on a site in the USA, to test how models do at a site they have never seen. I used the refined labels of SULAND_v2, which keeps the frames and splits and corrects missing and false boxes found in an audit of the first release (Lekhak et al., 2026), from its Hugging Face record at a fixed revision, also under CC BY 4.0 (Lekhak, 2026). The record states that the dataset “is intended for academic research”.

The dataset meets the brief’s rules: real photos rather than synthetic or AI-generated images, public, with a documented source and licence, and not used in an earlier project. It also shapes what the model can learn. The items are surrogates, not live mines, of two types only, filmed at two sites, so the generator can only learn what these two surrogates look like in these places. The surrogates themselves are not always true to shape: in the blind review I marked the shape wrong on 3 of the 8 real PFM-1 crops, because of both the surrogate’s shape and the view.

From Frames to Crops

The image archive is 16.7 GB, but it stores the photos uncompressed, so I read only the 9,911 frames that hold a usable box, 5.35 GB, by HTTP range requests, and checked each frame against the archive's CRC-32 checksum. Each labelled box was cut out as a square, widened by 15% of its longer side on each side to keep some ground around the item, moved inside the frame where it would run past an edge, and resized to 64 × 64 pixels. Boxes that touch the edge of the frame, where the item is cut off, or that are under 32 pixels on their longer side were dropped, which kept 10,397 of the 11,623 boxes in the three sets I used (89.5%; Table 1). The size of 64 pixels was set by a timing test, at about half a minute per pass over the training crops on a two-core processor. For the classifier check, each frame with a kept box also gave one square of the scene with no mine in it, mostly ground (Figure 1). The notebook ships the crops with their checksums, and it can rebuild them from the archive and confirm they are identical.

Table 1

Crops by Set and Type

Set	PFM-1	PMA-2	Background	Videos
Italy, training	2,981	2,909	5,675	34
Italy, test	468	465	902	6
USA	2,287	1,287	3,334	10

Note. From notebook Sections 2.2 to 2.4. The generator learned from the Italian training crops only; the test and US crops were held out to judge it.

Data Quality

Three checks shaped the evaluation. First, the items are small in their frames: in Italy the median kept box is 99 to 122 pixels on its longer side, and a tenth are under 40 to 47, so many crops are enlarged from few pixels and look soft. Second, the training crops repeat. Frames of one video show the same few items from slightly different points of view, the five largest of the

34 training videos give 27% of the training crops, and by a 64-bit difference hash, 24% of the training crops have a near-duplicate among the others. No test or US crop has one among the training crops, as expected of sets made by video and site, so I compared the generated images with the test crops, not the training crops, when checking for copies. Third, the surrogates were painted differently at the two sites: the Italian ones are green, and the US ones lighter and greyer, above all the PMA-2, so a model trained on the Italian crops has seen neither these colours nor this ground.

Figure 1

Real Crops of Each Type and of Ground, From Italy and the USA



Note. From notebook Section 2.4. Each crop is 64 × 64 pixels. Crops from SULAND (Lekhak, 2026; Vivoli et al., 2024), CC BY 4.0.

Model Design and Training Approach

Why a Class-Conditional GAN

The brief allows a GAN, a variational autoencoder (VAE) or a Transformer-based generator. Images call for a GAN or a VAE, and I chose a GAN, which tends to give sharper images than a VAE of the same size but is harder to train. Making it class-conditional lets me ask for either type, which the uses above need and which makes the outputs testable: each generated image comes with the type it should show. A GAN at 32 × 32 pixels would have trained faster, but would have kept too little detail to judge shape.

The Generator and the Discriminator

The layout follows the deep convolutional GAN, DCGAN (Radford et al., 2016).

- **Generator.** It takes 100 random numbers and the one-hot mine type, (1, 0) for a PFM-1 and (0, 1) for a PMA-2. A transposed convolution spreads these 102 inputs over a 4 × 4 grid of 256 channels, and four more each double the grid and halve the channels, to a 64 × 64 image with three colour channels. Every layer but the last is followed by batch normalisation, which keeps the activations on a steady scale while the two networks keep changing each other's targets (Ioffe & Szegedy, 2015), and a ReLU; the last uses tanh, so the outputs fall between −1 and 1, like the scaled crops. It has 1,108,416 weights.
- **Discriminator.** It takes an image and its type, given as two extra input channels that are constant over the image, so that it judges whether the image is a convincing crop *of that type*. Strided convolutions halve the grid four times, with leaky ReLU and batch normalisation on every layer but the first, and a last convolution gives one score. It has 695,680 weights.

- **Width.** Both networks are half the width of the standard DCGAN. In a timing test on the two-core workspace, a training step took 0.30 seconds at this width against 1.08 at the standard one, the difference between about 20 minutes of training and more than an hour.

Training

Each step trains the two networks in turn on a batch of 64 training crops. The discriminator learns, through binary cross-entropy, to score real crops as real and generated images of the same types as generated; the generator learns to make the discriminator score its images as real, the non-saturating form of the GAN loss, which gives it strong gradients early on, when its images are easy to reject (Goodfellow et al., 2014). Both use Adam (Kingma & Ba, 2015) at a learning rate of 0.0002 with $\beta1 = 0.5$, and the weights start from a normal distribution with a standard deviation of 0.02, the DCGAN settings (Radford et al., 2016). Each crop is flipped left to right at random half the time, since an item seen from the other side is still an item of its type. Every random step is seeded, PyTorch uses deterministic algorithms, and training runs on the processor with two threads, so a rerun on the same computer gives the same images and numbers.

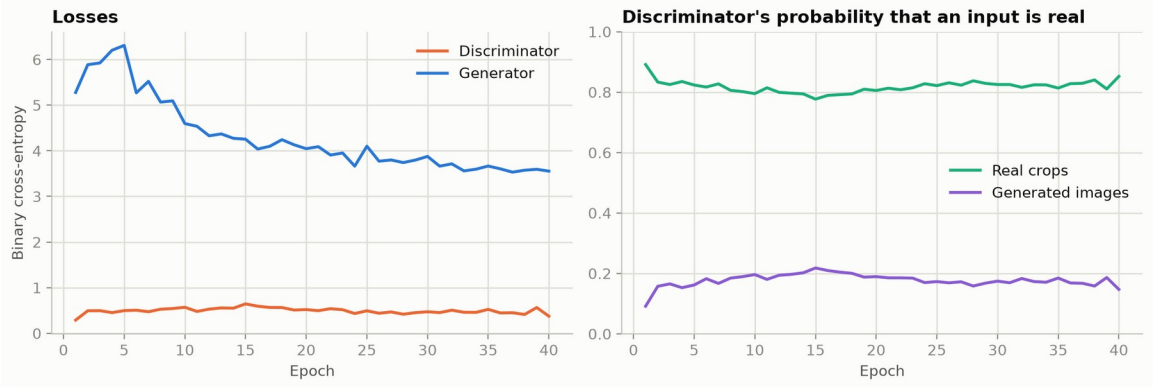
Short pilot runs on the training crops settled these settings. The standard layers, trained for 40 epochs, gave images in which both types could be recognised. Two common changes did no better: upsampling by nearest neighbour followed by a convolution, a remedy for the chequered pattern that transposed convolutions can leave (Odena et al., 2016), gave blotchier images and took about 20% longer per epoch, and spectral normalisation in the discriminator gave no recognisable shapes in 20 epochs.

The 40 epochs took about 15 minutes, and neither network won (Figure 2). After the first epoch the discriminator gave real crops an average probability of being real of 0.78 to 0.85, and

generated images 0.15 to 0.22, so it told them apart most of the time but was never certain, and the generator kept receiving useful gradients. The generator's loss rose in the first five epochs, while the discriminator learned faster, and then fell, with small rises, to 3.55 by the last epoch, so its images grew harder to reject. Images made from the same random inputs through training show blurred blobs after one epoch, a chequered pattern after five, ground with a coloured patch after ten, and recognisable items of both types by thirty. Each input gives a different scene, so the generator did not collapse onto a few images.

Figure 2

Losses and the Discriminator's Scores Through Training



Note. From notebook Section 5.2. Left: the binary cross-entropy of each network, averaged over each epoch. Right: the discriminator's average probability that real crops, and generated images, are real.

Output Evaluation and Interpretation

The Samples

Every generated image I used was stored the way the real crops are, as a JPEG at quality 95 and read back, so that no check could tell real from generated images by compression alone. Beside real training crops, the samples show that the generator tells the two types apart: the PFM-1 samples show green items and the PMA-2 samples round ones, on gravel, leaves, grass or

soil, and the grounds vary as much as the training crops' do (Figure 3). But the items are less clear than in the real crops. Their outlines are soft and uneven, some samples show a faint item or none, and at full size the images carry a fine chequered pattern that the real crops do not have.

Figure 3

Real Training Crops and Generated Samples of Each Type



Note. From notebook Section 6. Each panel shows 36 crops of 64 × 64 pixels. The generated samples come from new random inputs. Real crops from SULAND (Lekhak, 2026; Vivoli et al., 2024), CC BY 4.0.

Do They Copy the Training Crops?

A generator that had memorised its training crops would return near-copies of them. I found each sample's nearest training crop by the root-mean-square difference of their pixels, and compared it with the same distance for the real test crops from Italy, which come from other videos. The samples are no closer to the training crops than new real photos are: the median distance is 0.219 for the samples and 0.218 for the test crops, and none of the 128 samples is closer to a training crop than the closest test crop is. The closest samples are grey gravel scenes that reproduce the ground and light of a heavily repeated training scene, with a dark smudge where its small item lies. The generator reproduces the look of scenes it saw often, not individual photos.

What a Classifier Trained on Real Crops Makes of Them

A small convolutional network trained on the real training crops to sort crops into PFM-1, PMA-2 and background never saw a generated image, so it can judge whether a sample looks like its type. It classed 91% of the generated PFM-1 samples as PFM-1 and 75% of the PMA-2 samples as PMA-2, and called most of the rest background. Real test crops from Italy fare almost the same, at 93% and 73%. By this measure the type input is obeyed about as often as a real photo is recognised. The blind review below shows what this measure misses.

Can They Train a Detector?

If generated images are to stand in for real ones as training data, a model trained on them should work on real photos, the "train on synthetic, test on real" test (Esteban et al., 2017; Ravuri & Vinyals, 2019). I trained the same small classifier twice, once on the 5,890 real training crops and once on 5,890 generated images in the same numbers of each type, both with the same real background crops, and scored both on real crops from Italy's test videos and from the US site (Table 2). Since crops from one video share its items, light and ground, the 95% intervals come from resampling whole videos, and with 6 and 10 videos they are wide.

Table 2

Classifiers Trained on Real or on Generated Mines, Tested on Real and Generated Crops

Trained on	Tested on	Macro-F1 [95% interval]	Mines found [95% interval]	False alarms
Real mines	Italy, test videos	0.90 [0.76, 0.98]	0.84 [0.66, 0.97]	0.006
Real mines	USA, new site	0.43 [0.31, 0.55]	0.38 [0.24, 0.54]	0.003
Generated mines	Italy, test videos	0.26 [0.22, 0.28]	0.03 [0.00, 0.06]	0.000
Generated mines	USA, new site	0.22 [0.21, 0.22]	0.002 [0.00, 0.00]	0.000
Generated mines	Italy backgrounds, generated mines	0.99 [0.99, 1.00]	1.00 [1.00, 1.00]	0.000

Note. From notebook Section 7.2. Macro-F1 averages the F1 scores of PFM-1, PMA-2 and background. Mines found is the share of mine crops classed as either mine type; false alarms, the

share of background crops classed as a mine. The intervals come from 2,000 resamples of whole videos.

Trained on real crops, the classifier did well on new videos from the training site and poorly at the US site, where it found 38% of the mines: the drop at an unseen site that the US set was kept apart to measure. Trained on generated mines, it was of no use on real photos. It found 3% of the real mines from Italy and almost none from the USA, calling nearly every real mine background. Yet on new generated images it scored 0.99 and found every mine. Judged on generated data alone, it would look almost perfect.

Three further checks found out why (Table 3). In its training set every mine is generated and every background real, so any difference between generated and real images separates its classes as well as the items do: a shortcut (Geirhos et al., 2020).

Table 3

Why the Classifier Trained on Generated Mines Fails

Check	Trained on real mines	Trained on generated mines
Real Italian mines found, images blurred 4 × 4	81%	6%
Type right on real Italian mine crops, type alone	95%	84%
Type right on real US mine crops, type alone	62%	55%

Note. From notebook Section 7.2. The blur removes the generated images' chequered pattern. The type-alone classifiers learn only PFM-1 against PMA-2, so the difference between generated and real images cannot separate their classes. A separate classifier told every real crop and generated image it had not seen apart.

- **The chequered pattern is there, but it is not the cause.** The generated images' power spectrum peaks at half and a quarter of a cycle per pixel, 26 and 7 times the neighbouring frequencies, and the real crops show no peak. A 4 × 4 blur removes these frequencies, yet

trained and tested on blurred images, the classifier trained on generated mines still found only 6% of the real mines, against 81% for the one trained on real mines.

- **The type does transfer.** Trained only to tell PFM-1 from PMA-2, where both classes are generated and the shortcut cannot help, a classifier trained on generated mines got the type of 84% of the real Italian test crops right, against 95% when trained on real ones. It was as good on the PMA-2 (99%) but worse on the PFM-1 (69% against 90%).
- **Generated images are easy to spot.** A classifier trained to tell real crops from generated images, with both blurred, sorted every crop it had not seen correctly: all 933 real test crops from Italy and all 3,574 from the USA, and all 933 new generated images. This is a classifier two-sample test (Lopez-Paz & Oquab, 2017), and it says the two sets of images differ in ways a small network picks up.

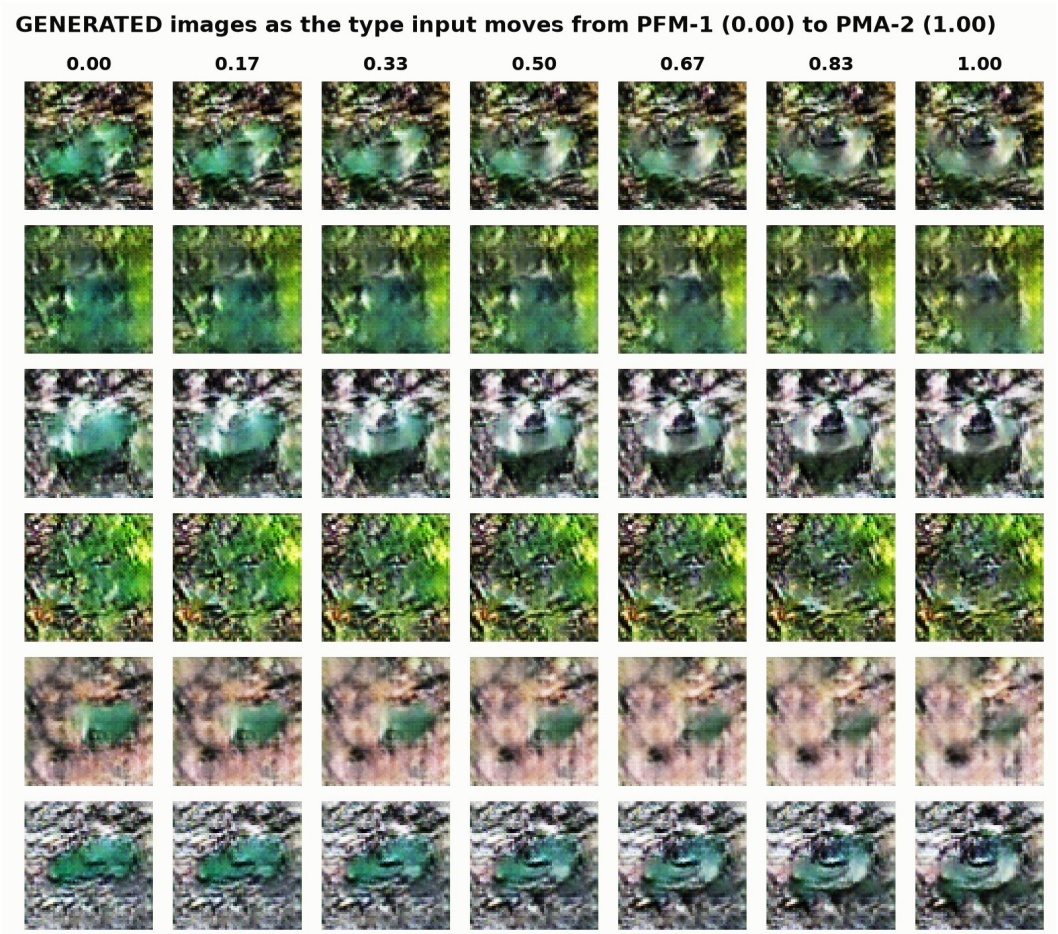
So the generated images hold much of what tells the two types apart, but they also differ from real photos, beyond the chequered pattern, in ways a small network picks up without fail. A detector trained on them beside real ground learns those differences instead of the mines, and only a test on real photos shows it.

Between the Two Types

Asked for a type halfway between PFM-1 and PMA-2, the generator does not refuse (Figure 4). Holding the random input fixed and moving the type input in seven steps, the item changes gradually from one type to the other in four of six rows, while the ground and light stay the same; in the middle steps it shows blends of the two outlines that no real mine has. The generator has learned one continuous range of items rather than two separate types, so it will draw an impossible item if asked, the most misleading kind of output for recognition training.

Figure 4

Generated Images as the Type Input Moves From PFM-1 to PMA-2



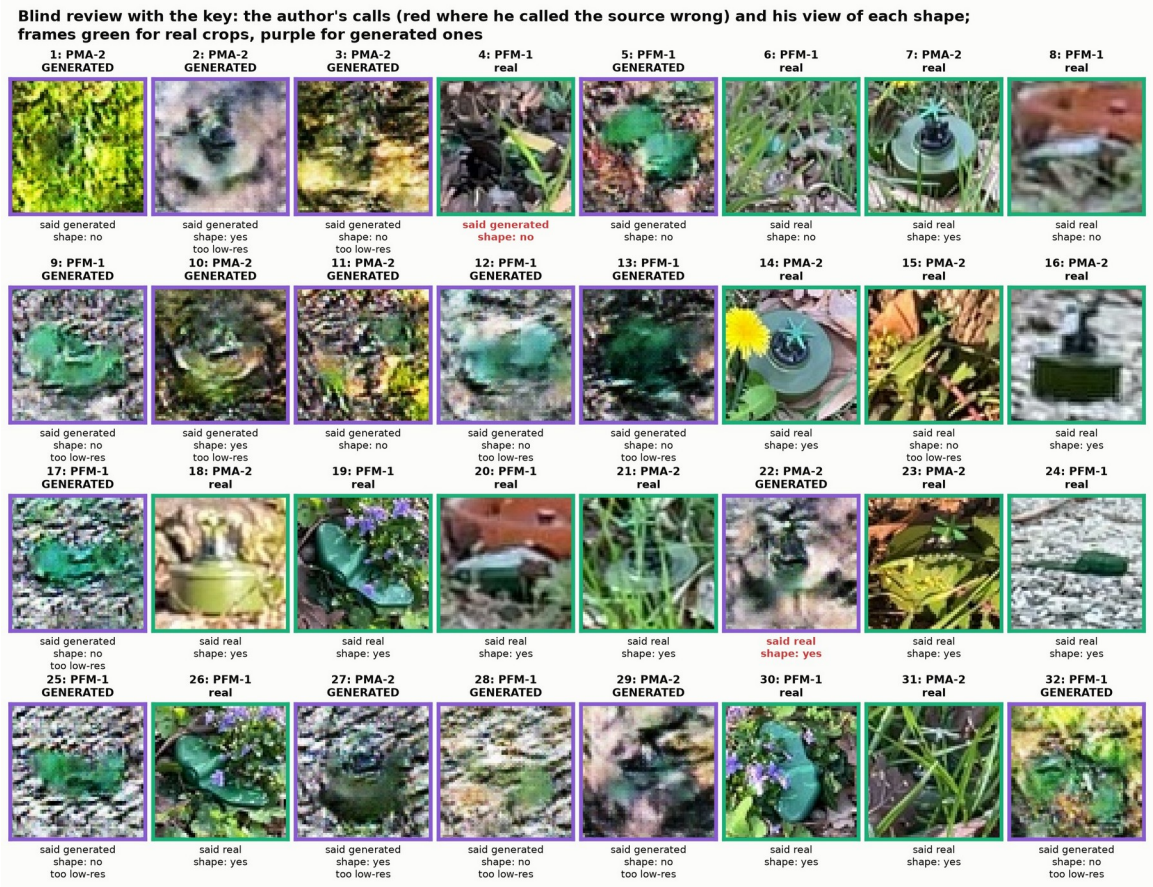
Note. From notebook Section 7.4. Each row uses one fixed random input; the columns move the type input from PFM-1 (0.00) to PMA-2 (1.00). All images are generated.

The Blind Review

To see the images through the eyes of their intended users, I reviewed 32 crops on one sheet, 16 real from Italy's test videos and 16 generated, 8 of each type, shuffled and numbered with the type shown and the source hidden. For each, I said whether it was real or generated and whether the item's shape is right for its type (Figure 5).

Figure 5

The Blind-Review Sheet With the Key



Note. From notebook Section 7.5. Frames are green for real crops and purple for generated ones; red text marks the two sources I called wrong.

I called the source right for 30 of the 32 crops (Table 4), 15 of the 16 real and 15 of the 16 generated; if every call were a coin toss, 30 or more right would come up about once in 8 million tries. I first set aside 13 crops as “all too low-res to even identify”, and 12 of them were generated, 12 of the 16 generated crops, against 1 of the 16 real ones. Every crop has the same 64 × 64 pixels, so what I read as low resolution is the generator’s lack of detail, not the pixel count. I judged the shape right in 12 of the 16 real crops, but in only 4 of the 16 generated ones, all

PMA-2: none of the 8 generated PFM-1s had the right shape. One generated PMA-2 passed as real, with the right shape, so the generator can produce a convincing item, but rarely.

This is the gap the classifier could not see. It classed 91% of the generated PFM-1 samples as PFM-1, yet none of the 8 on the sheet had the right shape. The classifier seems to accept a green item in the middle of the frame; a specialist looks for the shape, which is what matters when mines can be painted to camouflage.

Table 4

Blind Review of 16 Real and 16 Generated Crops

Crops	Source called right	Too low-res to identify	Shape right
Real PFM-1 (8)	7	0	5
Real PMA-2 (8)	8	1	7
Generated PFM-1 (8)	8	7	0
Generated PMA-2 (8)	7	5	4

Note. From notebook Section 7.5. My answers were given before I saw the key. A working file had shown the sources of two crops before the review; without those two, 28 of the other 30 sources were called right.

In Plain Terms

The generator makes images that look like mines of the requested type to a classifier and like the training scenes to the eye, without copying them. To an EOD specialist, nearly every one is recognisably generated, most are too indistinct to identify, and few show the right shape. As training data they teach a detector to spot generated images rather than mines, while looking almost perfect on generated tests.

Ethical Considerations and Responsible Use

Wrong Pictures Teach Wrong Shapes

The most direct risk is that someone learns what a mine looks like from a picture that is wrong. A generated image carries no guarantee of being true to the item, and this one draws

blends of two types that no real mine has (Figure 4) and gave the PFM-1 a wrong shape in every generated crop I reviewed. In recognition training or awareness material, a deminer, a survey team or a local person could carry that wrong shape into the field. Pictures that teach people what a mine looks like should be grounded in engineering data and be as accurate as possible, and with the current samples I would not use these images anywhere in mine action: much more time needs to be taken to train on many more samples before they could be viable. For that reason every generated image in the notebook and this report is labelled as generated, and none should be used to show anyone what a mine looks like unless an expert has checked it against engineering data.

False Confidence

The second risk is a detector that looks better than it is. Synthetic data used to make a dataset more diverse or representative carries a high risk of false confidence (Whitney & Norman, 2024), and Table 2 shows that risk in this project's own numbers: a classifier trained on generated mines scores 0.99 on generated test images and finds 3% of real mines. A team that tested a detector only on generated images, or on real photos from the training site, could field one that misses most mines, where a missed mine can cost a life. Any detector trained wholly or partly on generated images must therefore be tested on real photos from the site where it will be used, and judged by how many real mines it finds.

Misuse of Generated Images

A generated or altered photo could in principle be used as false evidence: a fake mine to suggest contamination, or an altered scene to make a suspected area look clear. I see these as very fringe potentials, which might happen only if the incentives in an environment were badly skewed. The people who get into mine action are not there to mislead people, and land release is

an evidence-based decision-making process (United Nations Mine Action Service [UNMAS], 2019): non-technical survey, technical survey and clearance test what a report claims before any land is cancelled, reduced or cleared, so a fake image would be caught before it became a catastrophic issue. Labelling generated images as generated, as here, keeps them from entering that evidence by mistake.

Data, Licence and Ownership

The crops and the generated images derive from SULAND's photos under CC BY 4.0, which allows adaptation with attribution, so the repository credits both the original release and the v2 labels and states the changes made: the items cut out and resized, and the labels turned into tables (Lekhak, 2026; Vivoli et al., 2024). The items are surrogates laid out for filming, not live mines, and the record states that the dataset is intended for academic research, which is how it is used here. I documented the data in the manner of a datasheet (Gebru et al., 2021), with its source, licence, sets and sites, so that its limits travel with it. The notebook and this report stay at the level of mine types and counts, with no technical detail of how the items work.

Who Is Served Well

A generator trained on these crops serves some places far better than others. It has learned two surrogate types, painted green, in Italian grass, leaves and gravel. At the US site, where the surrogates are lighter and the ground differs, even a classifier trained on real Italian crops found only 38% of the mines (Table 2), and the generator has seen nothing else. Images from it would represent neither the many other mine types nor the sand, rubble and other grounds where mines are found, so any use of them would favour settings like the training site, often without anyone noticing.

Limitations and Future Improvements

Limitations

- **Two surrogate types at two sites.** The generator knows only two surrogates, filmed in Italy, and the surrogates themselves are not always true to shape.
- **Few distinct items.** The 5,890 training crops come from 34 videos that show the same items again and again, and 24% have a near-duplicate, so the generator learned from far fewer distinct items than the count suggests.
- **Low resolution.** At 64 × 64 pixels, many crops are enlarged from small boxes and lose the detail of the shape, the feature that matters most.
- **One training run.** The generator was trained once, with one seed, so I cannot say how much its outputs would vary between runs, and on another computer the images and scores differ somewhat.
- **One reviewer.** The blind review rests on 32 crops judged by one EOD specialist, myself, who knew the project; a panel of reviewers and more crops would give firmer numbers.
- **A classifier as judge.** The classifier trained on real crops accepted shapes that a specialist rejected, so its acceptance rates measure resemblance to the training photos, not correct shape.
- **One way of using generated data.** I tested only generated mines beside real ground. Mixing generated and real mine crops, or generating the ground as well, might avoid the shortcut, and I did not test either.

Future Improvements

- **Start from a 3D model as the ground truth.** It would be interesting to build a really good 3D model of each item from engineering data, then place it under different

overlays: vegetation patterns obscuring it, different levels of obscuration, and different colour patterns, as if it were painted or hidden. The shape would then be right by construction, and the variation would come from the overlays, close to domain randomisation, which trains a detector on simulated images with randomised textures, lighting and positions so that it carries over to real photos (Tobin et al., 2017). Recent work on humanitarian demining goes this way: a detector trained on rendered 3D scenes of objects in vegetation, re-textured by a diffusion model, matched or exceeded one trained on a larger set of real images from a different site (Malizia et al., 2026).

- **More and better data.** Many more photos of each type, from more sites, more mine types and sharper images would all play a key role. This project has reinforced for me the value of proper data collection and example collection in the field. I would like to see a larger, properly cleaned dataset, and EOD teams need to start introducing a data capture strategy, so that we gain more insight into these things.
- **A different generator.** Diffusion models, which learn to remove noise from an image step by step, give high-quality images (Ho et al., 2020); the brief does not ask for them, but they would be the next generator to try, conditioned on the type or on a 3D render.
- **Test where it will be used.** Whatever generates the images, a detector trained on them should be judged on real photos from the site of use, and the images checked by a panel of specialists against engineering data before any use in training material.

Use of AI Tools

I used Claude (Anthropic) in a supporting role for this project. All code has been run on my computer.

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